Juke Manual

A Desktop Quick Reference

The Juke Project

Covers Juke 0.0.1

> *The lyrebird (Menura novaehollandiae) is one of nature's great mimics — it records the sounds of its world and replays them with uncanny fidelity, much as Juke records and replays your upstream interactions.*

Table of Contents

Preface

Test code spends an extraordinary amount of its life pretending. A unit test pretends an inventory service returned three widgets; an integration test pretends a payment gateway timed out; a UI test pretends the weather API is having a good day. Every one of those pretences is hand-written, drifts away from reality the moment the real system changes, and has to be maintained forever.

Juke replaces the pretending with *recording*. It captures what real upstream systems actually did, stores it in a portable archive, and replays it deterministically — so your tests run against real data with no mock code to write or own.

This book is a quick reference for developers and test engineers who use, or are evaluating, Juke. It covers the open-source **Community** distribution in full and points to where the **Enterprise** modules extend it.

Who This Book Is For

You should be comfortable with Java and Spring Boot. You do not need prior experience with mocking frameworks, service virtualization, or recording tools — Juke's model is explained from first principles in Chapter 3. If you write Playwright, Cypress, or Selenium tests against a Spring Boot backend, Part I will be immediately useful; if you operate Juke in CI, Part II is your desk reference.

How This Book Is Organized

The book is deliberately layered. **Part I** builds understanding from the top down: each chapter opens with the general idea and why it matters, then descends into nuance. Read it front to back the first time.

- **Chapter 1** makes the case for Juke: unit versus behavioral testing, why whole-system journeys matter, why AI-written unit suites can drift, and where Juke fits.
- **Chapter 2** gets a recording replaying in five minutes.
- **Chapter 3** explains the mental model — the interception seam, the modes, the recording.
- **Chapters 4–8** are the working core: annotations, modes, sessions, the recording format, and runtime fault injection.
- **Chapters 9–10** cover the newer reporting surface — functional coverage and session reports.

Part II is dense reference: the complete HTTP contract, every configuration key, and the storage SPI. Reach for it once the concepts in Part I are familiar.

Part III collects the sample applications, a troubleshooting catalog, and a glossary.

Conventions Used in This Book

The following typographic conventions are used:

Italic

New terms, filenames, and URLs.

`Constant width`

Code, annotations, configuration keys, command-line input, and the names of classes and methods.

`Constant width italic`

Text that should be replaced with a user-supplied value.

> **Note** — A note flags a useful aside or a point of clarification.

> **Tip** — A tip is a recommended practice or a shortcut.

> **Warning** — A warning calls out a sharp edge that can cost you time.

Using Code Examples

Every command, annotation, and configuration snippet in this book is copy-pasteable. Examples target Juke 0.0.1 on Java 25 and Spring Boot 3.5. Where a command differs between shells, both forms are given.

Part I — Understanding Juke

Chapter 1. Introduction to Juke

This chapter answers two questions: *what is Juke*, and *why would you choose it over the testing tools you already have*. The rest of the book assumes the answers.

1.1 What Juke Is

Juke is a **record-and-replay framework for Spring Boot services**. It sits between your service-layer beans and the upstream collaborators they call — other services, databases, REST endpoints, message brokers, anything reachable through a Spring-injected interface or `*Template`.

In **record** mode it captures every upstream interaction as a JSON document inside a portable ZIP archive. In **replay** mode it serves those documents back deterministically, so no upstream call ever leaves the host process. The result is a deterministic test fixture made of *real* data: UI tests, integration tests, and CI pipelines all see byte-identical responses run after run, regardless of what the underlying upstream system is doing.

At a high level, Juke does two things: it captures real upstream behavior, and it replays that behavior deterministically later. Everything else in this book — seams, modes, sessions, storage, and reports — is detail hanging off those two ideas.

1.2 Unit Testing and Behavioral Testing

The central testing tradeoff is simple: **unit tests give control**, while **behavioral tests give truth**.

Unit tests isolate one class or function at a time. They are fast, precise, and excellent for pinning down small rules, edge cases, and regressions in code you already understand. They fail close to the source of a defect, they are cheap to run in large numbers, and they make refactoring safer when the behavior is already well-specified.

Their weakness is that they depend on *authored substitutes* for the rest of the system: mocks, stubs, fixtures, and expectations. That makes them only as good as the author's understanding of the surrounding behavior. If the mental model is incomplete, the test is incomplete in the same way. Unit tests are therefore strongest as a guardrail around local logic, and weakest as proof that the whole feature works as a user experiences it.

Behavioral tests start from the opposite end. They drive the system through an externally visible flow and judge success by the outcome a user or caller actually sees. Their great strength is independence: they do not need to know how the system is implemented internally to tell you whether the promised behavior is present. They are the best tool for catching mismatches at seams — between UI and server, controller and service, service and upstream, or one runtime environment and another.

Their weakness is operational. Real behavioral tests are often slow, brittle, hard to set up, and expensive to keep reliable because they depend on live upstreams, mutable test data, network timing, and shared

environments. In practice, teams often settle for many unit tests and only a small number of true end-to-end journeys, even though the journeys are the tests that answer the most important question: *does the feature actually work?*

1.3 How Juke Bridges the Gap

Juke exists because teams should not have to choose between the *speed and control* of unit tests and the *truthfulness* of behavioral tests.

It bridges that gap by splitting the problem in two:

1. **Record once against reality.** Drive a real use case through the running system and capture what the upstreams actually returned.
2. **Replay many times deterministically.** Re-run that same use case against the captured interactions, with no live upstream dependency.

That combination preserves what matters from both worlds.

- From **behavioral testing**, Juke keeps the real code path, real data, and a validator anchored to what a user actually drove.
- From **unit-style testing**, Juke keeps repeatability, speed, isolation from unstable external systems, and suitability for CI.

The result is a test that is not built around a hand-written fake, yet is also not hostage to the unpredictability of live environments. Recordings turn real behavior into a deterministic asset. That is Juke's core idea.

> **Note** — Put plainly: Mockito tests your code against *your assumptions*, WireMock tests it against *a stub you maintain*, and Juke tests it against *what the real system actually did*. A passing Juke journey is therefore stronger evidence than a passing unit test, while still being stable enough to run repeatedly as part of normal development.

1.4 Why the Whole System Matters

Modern applications do not keep their logic neatly on one side of the boundary. Validation, retries, feature flags, formatting, optimistic updates, loading states, fallback behavior, permission checks, and error handling are often distributed across browser code and server code together. The user experiences the *composition* of those pieces, not either side in isolation.

That is why testing the UI and the server as separate concerns is increasingly insufficient. A server unit test can prove that an endpoint returns a field; a UI unit test can prove that a component renders a field; neither proves that the right data crosses the boundary, at the right time, in the right shape, under the right real-world conditions.

Whole-system behavioral testing answers a different question: not "did the UI logic pass?" and not "did the server logic pass?" but **"did the user journey work?"** That is the level at which many expensive bugs actually live.

This does not make unit tests obsolete. It changes their role. Unit tests are best for local correctness; whole-system behavioral tests are best for feature correctness. You need both — but if forced to choose which one should carry the stronger claim about whether the product works, the answer is the test that drives the product the way its users do.

1.5 The AI-Test Trap

AI coding assistants make this structural weakness acute. A model that writes both the implementation *and* the test does so from the same internal representation of what the code should do. The test then passes not because the behavior is correct, but because the assertion was written to match the behavior the model already had in mind. **Green means "the code does what I thought," which is circular.** The more capable the model, the more convincingly self-consistent — and self-confirming — the test suite becomes.

Every unit test has two halves: the code that exercises the system, and the *validator* that says what "correct" means. When both halves come from the same source, you are not testing the code against an independent standard; you are photographing the current behavior at high resolution. That photograph can look exactly like a passing test suite while hiding real bugs.

Thought experiment 1 — the missing mapping. An AI writes a `PricingService` that applies a 10% discount for premium users, then writes a unit test that mocks the premium flag and asserts the discount is applied. Green. But when you record a real premium user's session, the real code runs against real data with an unexpected coding — and the discount never appears in the recorded response. The bug is visible the moment you capture it. Unit tests: green. Real behavior: wrong.

Thought experiment 2 — drift masked by test maintenance. Later, the AI is asked to add caching to `PricingService`. Its context is narrowly focused on making the cache work. To do that cleanly, it adjusts how the discount calculation is invoked. The existing discount test now fails — not because the discount is broken, but because the invocation path changed. The AI fixes the test to match the new path. Green again. What it didn't reason about: the old invocation path had a side effect that fee-audit logging depended on. Audit records for premium discounts now silently disappear. No test ever asserted on audit-log completeness — the original author "knew" the discount method would always be called directly, so they never thought to guard it.

This is **model drift masked by test maintenance**. The mental model shifts incrementally with each task, tests are updated to match the new model, and the suite stays green while the system's actual contract with its users quietly erodes. Three properties make it structural:

1. **Tests are updated, not just added** — passing green is restored, not earned.
2. **The validator is the author** — no independent behavioral anchor exists to veto the drift.

3. **Each step is locally reasonable** — no single change looks wrong; only the cumulative delta against real behavior reveals it.

Why code review doesn't catch it. The natural safety net — a human reviewer spotting the drift — erodes under AI-assisted velocity. Each individual change is small, locally reasonable, and arrives with green tests as evidence of correctness. A reviewer assessing a caching change sees clean code, a passing build, and a test suite that grew by two tests. They are not in a position to reconstruct the cumulative delta against the system's original behavioral contract from that diff alone. At scale, review becomes an audit of style and structure, not a check on semantic correctness. The logical drift is spread across dozens of small PRs, invisible in any single one, and only visible in aggregate — which no reviewer ever reads as an aggregate.

This is not a failure of discipline or attention. It is a structural consequence of high-velocity AI-assisted development: the rate of change outpaces the bandwidth of human comprehension applied diff-by-diff. The only gate that scales with that velocity is an independent validator that never reads the diffs at all — it just asks whether the system still delivers what it promised.

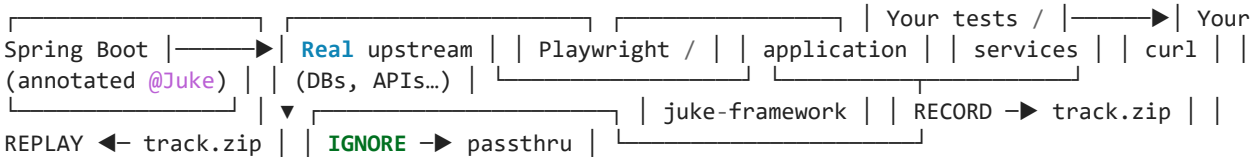
A behavioral journey is that validator. It is an independent flow driven end-to-end, asserting on a user-visible outcome. Nobody updated it when the implementation changed — it still asserts what the system *promised*. That is why it can disagree with the code, and disagreement is exactly the signal you want.

The defensible validation gate is therefore:

> *Full build green* + *behavioral journeys passing*

Unit tests guard against regressions in behavior you already understand. The behavioral journey tells you whether the behavior was ever right to begin with — and whether it has quietly drifted since.

1.6 Where Juke Fits



Juke does not replace Playwright, JUnit, or your CI system — it sits beneath them, supplying the server-side data layer they have always had to fake.

1.7 The Juke Distribution

The Community distribution is four Apache 2.0 jars, plus an optional fifth:

Module	Role

`juke-framework`	The engine — annotations, proxies, the storage SPI, the session registry.
`juke-remix-rest-service`	The `/service/*` HTTP control surface — record, replay, sessions, reports.
`juke-plugin-api`	DTOs for the plugin contract.
`juke-plugin-sdk`	A Spring Boot starter for writing plugins.
`juke-coverage`	*Optional.* Functional code-coverage capture (Chapter 9).

The Enterprise modules — `juke-scenario-service` (JPA persistence, database-backed storage), `juke-admin-server`, and the `juke-admin-ui` SPA — are strict additions; nothing in Community changes when you bring them in.

1.8 How Juke Compares

Capability	Juke	Mockito	WireMock	Spring Cloud Contract
Zero hand-written fakes	✓	✗	✗	✗
Records from live traffic	✓	✗	⚠ manual	✗
Exercises the real code path	✓	✗	✓	✓
Deterministic sequenced replay	✓	⚠	⚠	✗
Concurrent session isolation	✓	N/A	✗	✗
Runtime fault injection, no restart	✓	⚠	✓	✗

Portable archive of recordings	✓	✗	✓	✓

Chapter 2. Getting Started

This chapter takes a Spring Boot application from "no Juke" to "replaying a recording" in about five minutes. The concepts behind each step are deferred to Chapter 3 — here the goal is a working result.

2.1 Prerequisites

- Java 25
- Maven 3.9 or newer
- An existing Spring Boot 3.x application, or one of the bundled samples under `juke-samples/` (see Appendix A)

2.2 Add the Dependencies

```
<dependency> <groupId>org.juke.harnesss</groupId> <artifactId>juke-framework</artifactId>
<version>0.0.1-SNAPSHOT</version> </dependency> <dependency> <groupId>org.juke.harnesss</groupId>
<artifactId>juke-remix-rest-service</artifactId> <version>0.0.1-SNAPSHOT</version> </dependency>
```

`juke-framework` carries the engine; `juke-remix-rest-service` adds the `/service/*` control surface. Open the component scan so both are found:

```
@SpringBootApplication @ComponentScan(basePackages = {"org.juke.framework", "org.juke.remix",
"com.yourcompany"}) public class YourApplication { … }
```

2.3 Turn Juke On

Juke is **off by default**. A single YAML key opts an environment in:

```
juke: enabled: true # master switch path: /tmp/juke # directory holding recording ZIPs
```

> **Warning** — When `juke.enabled` is absent or `false`, the framework wires *nothing*: no proxies, no session beans, no `/service/*` controllers. Keep the key out of `application-prod.yml` so production stays Juke-free.

2.4 Annotate One Collaborator

```
@Service public class CheckoutService { @Juke private InventoryClient inventory; // ◀ this
dependency is intercepted public Order place(String sku, int qty) { return inventory.reserve(sku,
qty); } }
```

2.5 Record, Then Replay

Boot the app, then drive the lifecycle over REST — no restarts:

```
# Record curl "http://localhost:8080/service/record/start?track=morning-flow" curl
"http://localhost:8080/checkout?sku=A1" curl "http://localhost:8080/service/record/end" -o
morning-flow.zip # Replay, scoped to one test session curl -c cookies.txt
"http://localhost:8080/service/session/start?track=morning-flow" curl -b cookies.txt
"http://localhost:8080/checkout?sku=anything" curl -b cookies.txt
"http://localhost:8080/service/session/stop"
```

The replayed call returns the recorded response regardless of the `sku` you send. That is the deterministic guarantee — and the rest of Part I explains how it works.

Chapter 3. How Juke Works

Chapter 2 produced a result; this chapter supplies the model behind it. Once you hold the model, every endpoint, mode, and file in the later chapters has an obvious place to hang.

3.1 The Interception Seam

Juke does its work at a single point: the **`@Juke`-annotated injection point**. When Spring finishes wiring a bean, Juke's `BeanPostProcessor` replaces the annotated field's value with a *proxy* of the same type. The proxy implements the field's interface (or subclasses its class) and intercepts every method call.

Nothing else in the application is aware of this. The controller, the service logic, the serialization layer — all of it runs unchanged and unmodified. The *only* thing that changes is what happens when execution crosses the seam.

3.2 Record, Replay, Ignore

At the seam, the proxy consults the current **mode** and acts:

Mode	At the seam
`record`	Call the real upstream; capture the response, arguments, and type into the active ZIP; return the real response.
`replay`	Do *not* call the upstream; read the next recorded response from the ZIP, deserialize it, and return it.
`ignore`	Call the real upstream; capture nothing. Pure pass-through.

Modes are covered fully in Chapter 5. The key idea: the application code is identical in all three; only the proxy's behaviour at the seam differs.

3.3 Anatomy of an Interception

Each intercepted call produces a recording **identifier**:

```
<TypeName>.<methodName>.<sequence>
```

- *TypeName* — the interface simple name for `@Juke` on a field; the concrete class simple name for `@Juke` on a class.

- *methodName* — the Java method name; overloads are disambiguated by a parameter-signature hash.
- *sequence* — a 1-based per-`(type, method)` counter.

Three calls to `IGreetingsService.greeting(...)` in record mode produce `IGreetingsService.greeting.1`, `.2`, and `.3`. In replay mode the counter resets at session start, so the same three calls return the same three responses in the same order.

> **Note** — Replay is *sequenced*, not *keyed by argument*. The fourth call to a method that was recorded three times has nothing to return — see "JukeReplayNotFoundException" in Appendix B.

3.4 The Whole Picture

```
@Juke field mode = record mode = replay ──────────── ──────────── ──────────── proxy.call() ──▶
real upstream ──▶ ZIP proxy.call() ──▶ ZIP │ │ └─ response returned └─ recorded response to caller
returned to caller
```

Record once against the real world; replay forever against the archive. The remaining chapters add precision: which beans (Chapter 4), which mode and how it is chosen (Chapter 5), how concurrent tests stay isolated (Chapter 6), what the archive contains (Chapter 7).

Chapter 4. The Annotation Surface

Three annotations cover virtually every Spring injection pattern, and a fourth opts individual members out. This chapter is the working reference for all four.

4.1 `@Juke`

The workhorse. Its targets are `TYPE`, `FIELD`, `METHOD`, and `PARAMETER`; retention is `RUNTIME`.

On a field whose type is an interface — Juke installs a JDK dynamic proxy:

```
@Service public class OrderService { @Juke private InventoryClient inventory; @Juke private
PricingClient pricing; }
```

On a field whose type is a concrete class (e.g. a Spring `RestTemplate`) — Juke installs a CGLIB subclass that delegates to the real, fully-configured bean, so the wrapper is assignable back to the field. Two optional attributes apply to the concrete case:

```
@Autowired @Juke(name = "shipping", excludeMethods = {"setMessageConverters"}) private
RestTemplate restTemplate;
```

`name` overrides the recording-identity prefix (defaulting to the type's simple name) so two beans of the same type don't collide; `excludeMethods` skips config/builder methods you don't want recorded. A `final` concrete class can't be subclassed — type the field as the interface it implements instead. Per-session cookie replay currently applies to interface-typed fields only.

On a class — Juke installs a CGLIB subclass proxy, and *every* public method of the class and of every interface it implements is intercepted:

```
@Juke @Service public class FulfillmentService implements Shippable, Trackable { … }
```

Recordings for a class-level `@Juke` are keyed by the **concrete class name** (`FulfillmentService.ship.1`), not the interface name. `Object.equals`, `hashCode`, and `toString` are never intercepted.

On a method or parameter — Juke wraps the parameter before the method body runs, or wraps the return value after it returns.

The optional `value()` attribute pins this annotation to a mode regardless of the global setting:

```
@Juke("ignore") private DiagnosticsClient diagnostics; // never recorded — always pass-through
```

Accepted values: `juke` (default — follow the global mode), `record`, `replay`, `ignore`, `none`, `disable`. The `autoWrap()` attribute (default `true`) disables wrapping when set `false`.

4.2 `@JukeController`

`@JukeController` marks a Spring `@RestController` for controller-level AOP advice: request/response logging in MDC tags, optional finding capture, and cookie-bound session resolution. Most applications

never apply it directly — it is wiring for the bundle-backed session flow (Chapter 6).

4.3 `@JukeIgnorable`

When `@Juke` is on a class, every public method is wrapped. `@JukeIgnorable` opts one member out:

```
@Juke @Service public class OrderService { public Order place(String sku) { … } // recorded
@JukeIgnorable public void warmCache() { … } // never recorded }
```

It also drives field-level comparison rules. Its `IgnoreStrategy` has two values:

Strategy	Behaviour
`ALWAYS` *(default)*	Skip this field in every comparison — right for generated IDs and UUIDs.
`NOT_NULL`	Skip the value diff, but still flag a null-vs-non-null mismatch — right for timestamps.

Chapter 5. Operating Modes

A mode decides what the proxy does at the seam. This chapter covers the five modes, the three ways to set the global mode, and the precedence rule that lets per-session replay coexist with a global mode.

5.1 The Five Modes

Mode	Behaviour
`juke`	Default. Follows whatever mode is active elsewhere.
`record`	Calls hit the real upstream; responses, arguments, and type metadata are written to the active ZIP.
`replay`	Calls are intercepted before the upstream; recorded JSON is read back and returned.
`ignore`	Pass-through. Calls hit the real upstream; nothing is recorded.
`disable`	Master kill-switch. No interception at all — equivalent to removing every `@Juke`.

5.2 Setting the Global Mode

With `juke.enabled: true`, the global mode can be set three ways, later winning:

1. **REST at runtime (recommended).** `GET /service/record/start?track=name` flips to record; `GET /service/replay/start?track=name` flips to replay. No restart.
2. **YAML.** `juke.mode: record` in `application.yml`.
3. **System property.** `-Djuke=record` (a legacy alias; YAML is preferred).

5.3 Per-Session Versus Global

A Juke proxy checks for an active **session cookie before** it consults the global mode. The two mechanisms are layered, not exclusive:

Caller	Behaviour

Carries a valid `JUKE_SESSION` cookie	Replay against the session's track. *Global mode is irrelevant.*
No cookie, global mode `ignore`/`none`	Pass-through.
No cookie, global mode `record`	Capture into the active track.
No cookie, global mode `replay`	Replay against the globally-pinned track.
Any caller, global mode `disable`	Pass-through. The kill-switch wins.

This is why a test session can replay locally while production traffic on the same JVM stays pass-through.

5.4 Argument Validation

On replay, Juke compares each call's incoming arguments against the recorded arguments. `juke.args-validation` controls the response:

Value	On mismatch
`off`	Nothing.
`warn` *(default)*	Logs `INPUT MISMATCH […]`; the test continues.
`strict`	Throws `JukeReplayMismatchException` (surfaced as HTTP 500).

Use `strict` in gate jobs to turn silent input drift into a loud failure.

Chapter 6. Sessions and Per-Track Replay

A process-wide mode flip serves one consumer at a time. The session API gives every test worker or browser tab its own track, in parallel, against one JVM.

6.1 Why Sessions Exist

Three Playwright workers running in parallel each need a different track — `happy-path`, `cancel-flow`, `retry-after-timeout`. Without sessions you would have to `replay/start` between every worker's request, serializing the suite. With sessions, each worker calls `/service/session/start?track=…` once and receives a `JUKE_SESSION` cookie; the framework's cookie filter resolves that cookie to the right track on every subsequent request.

6.2 The Session Lifecycle

```
curl -c jc.txt "http://localhost:8080/service/session/start?track=happy-path" curl -b jc.txt
"http://localhost:8080/checkout?sku=A1" # replays happy-path curl -b jc.txt
"http://localhost:8080/service/session/stop"
```

`session/start` accepts an optional `description` query parameter that is stored with the session and surfaces in its report (Chapter 10). Each browser context maps 1:1 to a Juke session — in Playwright, set the `JUKE_SESSION` cookie on the context before navigating.

6.3 Status Visibility

`GET /service/sessions` returns one row per active session, including `lastCall` (the recording entry most recently served, rendered `<SimpleType>.<method>.<sequence>`) and `percentComplete`, a coarse 0–100 progress signal computed across every entry in the recording. It is built for live status grids.

When a session is stopped it moves into a per-track **history**, which the report endpoint of Chapter 10 reads.

Chapter 7. The Recording Format

A Juke recording is an ordinary ZIP archive of JSON. There is no proprietary format and no schema service — you can open one in any unzip tool.

7.1 The ZIP Layout

```
happy-path.zip ├── IGreetingsService.greeting.1.json ◀ response body ├──
IGreetingsService.greeting.1.args.json ◀ input arguments (sidecar) ├──
IGreetingsService.greeting.1.type.json ◀ runtime return type (sidecar) ├──
IGreetingsService.greeting.2.json │ … ├── juke.json ◀ class-metadata catalog ├──
juke-mappings.json ◀ short-name → FQN lookup └── juke-metadata.json ◀ track label + recordedAt
```

7.2 Three Files per Call

- `*.json` — the response, as Jackson serialized it. Replay returns this as the call's return value.
- `*.args.json` — the input arguments captured at record time. Replay compares incoming arguments against this sidecar (see §5.4).
- `*.type.json` — the runtime concrete return type, written when it differs from the declared type. Replay uses it to deserialize correctly.

7.3 Track Metadata

`juke-metadata.json` records the track's human-readable **label** (supplied via `record/start?…&label=…`) and the timestamp it was recorded. The session report (Chapter 10) reads it so a report can be titled meaningfully.

7.4 Why It Is Portable

Plain-text JSON, kilobytes per call, in a single ZIP. Commit recordings to your repository, attach them to bug reports, or share them between teams — no external dependency travels with the file.

Chapter 8. Remix: Runtime Fault Injection

Replay reproduces what the upstream *did*. Remix makes the upstream *misbehave* — on demand, on one specific call, against a recording you already trust.

8.1 Why Remix Exists

The hardest paths to test are the ones where an upstream is wrong in a specific, controlled way: a service slow exactly once, an exception thrown on exactly the second call. In unit tests these are trivial but unreal — the mock bypasses your retry and circuit-breaker code. In integration and Playwright tests they are nearly impossible without heavyweight chaos tooling. The result is suites that only ever exercise the happy path.

Remix injects precisely the failure you want, on precisely the call you want. The test stays a clean black-box scenario; the upstream misbehaves once, then recovers.

8.2 Inject an Exception

```
curl "http://localhost:8080/service/remix/exceptionSchedule?\
classAndMethodSequence=IGreetingsService.greeting.1&\
exception=IOException&exceptionMessage=Simulated+failure"
```

The next replay of that entry throws; the call after returns the recorded payload normally. Schedules are one-shot unless re-armed. This exercises retry catch-blocks, error-banner UX, and logging completeness.

8.3 Inject a Delay

```
curl "http://localhost:8080/service/remix/delaySchedule?\
classAndMethodSequence=OrderService.place.3&waitTimeInMS=10000"
```

A delay longer than a configured client timeout drives the timeout branch — code that is otherwise almost never executed. It also exercises loading-state UI and user-perceived-latency handling.

8.4 A Test Pattern

To validate "the UI recovers from a single transient failure": load a happy-path recording, schedule a one-shot exception on `Service.call.1`, drive the UI action, and assert the user sees a brief "retrying…" indicator followed by the recorded success. No mocks, no service virtualization, no real network failure — and your production retry code actually runs.

Chapter 9. Functional Test Coverage

Coverage tools answer "did my *unit tests* touch this line?" Juke can answer a more valuable question: "do my *real user-journey tests* exercise this code?" This chapter covers the optional `juke-coverage` module.

9.1 Use-Case Coverage Versus Unit Coverage

Unit-test coverage is measured against unit tests, which mock away most collaborators. **Functional** (use-case) coverage is measured while Playwright drives real flows through a Juke-replaying server. The *gap* between the two numbers is diagnostic: it reveals undertested workflows and dead code that no real journey reaches.

9.2 Server Coverage

`juke-coverage` reads the JaCoCo agent **in-process**. Start the server with the agent attached:

```
-javaagent:jacoco-agent.jar=output=none
```

Then `GET /service/coverage/server` pulls the live execution data, analyzes the application-under-test's classes, renders a drill-down HTML report, and returns a summary:

```
{ "available": true, "passed": true, "tool": "JaCoCo", "instruction": 84.2, "branch": 71.0,
"line": 86.5, "analyzedClasses": 4, "excludedSeams": ["com.example.IGreetingsService ->
com.example.GreetingServiceImpl"], "reportUrl": "/coverage/server/index.html" }
```

Because the agent accumulates over the JVM's lifetime, every replay session adds to the same figure — multi-run aggregation is automatic.

9.3 Excluding `@Juke`-Mocked Code

In replay mode, a `@Juke`-mocked implementation never executes — counting it would unfairly depress the result. Juke records every implementation it displaces with a proxy in a runtime registry (`JukeMockRegistry`), and the coverage analyzer skips exactly those classes. The developer never names an implementation class; the exclusion is a byproduct of the proxying Juke already does.

9.4 UI Coverage

Front-end coverage is produced outside the JVM. The SPA is built instrumented (`vite-plugin-istanbul`), Playwright harvests `window.__coverage__` after each test, and `nyc` renders a report. `GET /service/coverage/ui` reads that report's summary:

```
{ "available": true, "passed": true, "tool": "nyc/Istanbul", "lines": 78.0, "statements": 77.4,
"functions": 80.0, "branches": 62.5, "reportUrl": "/coverage/ui/index.html" }
```

9.5 The Combined Endpoint

`GET /service/coverage` returns both halves in a single payload — useful for dashboards and one-call CI gates:

```
{ "server": { "available": true, "passed": true, "tool": "JaCoCo", … }, "ui": { "available": true,
"passed": true, "tool": "nyc/Istanbul", … }, "passed": true, "generatedAt": "2026-05-20T10:00:00Z"
}
```

Each half carries its own `available` and `passed` flags; the top-level `passed` is `true` only when both halves are. If only the server half is attached (no Playwright run yet) the UI half reports `available: false` and the top-level `passed` reflects only the half that gates.

9.6 Thresholds and the `passed` Flag

Every summary carries a `passed` boolean. By default it is `true` (no thresholds set). Configure minimums in `application.yml` to make it gate:

```
juke: coverage: threshold: server: line: 80 branch: 60 ui: lines: 70 branches: 55
```

When a metric falls short, `passed` flips to `false` and `message` names every metric that missed and by how much. A CI script can fail the build on a single `jq` call without parsing percentages:

```
curl -sf http://localhost:8080/service/coverage | jq -e '.passed'
```

`jq -e` exits non-zero when `passed` is `false`, so the call drops straight into a pipeline step.

9.7 Opt-In, Production-Safe

The whole feature is gated by `juke.coverage.enabled` (default `false`) and, for server coverage, the presence of the `-javaagent`. A production deployment sets neither and pays nothing — the coverage beans are never instantiated.

Chapter 10. Session Reports

A single replay session is interesting; a suite of them is a *result*. The report endpoint consolidates every completed session for a track into one document.

10.1 The Report Endpoint

`GET /service/recording/report?track={name}` returns pretty-printed JSON:

```
{ "track": "demo", "label": "Greetings smoke test", "recordedAt": "2026-05-19T…", "sessions": [ {
"sessionId": "…", "description": "Normal replay", "callCount": 2, "overallStatus": "COMPLETED",
"calls": [ { "sequence": 1, "method": "greeting", "recordedArguments": ["Alice"],
"actualArguments": ["Alice"], "inputMatched": true } ] } ] }
```

10.2 Labels and Descriptions

Two optional query parameters feed the report:

- `record/start?track=…&label=…` — the track label, stored in `juke-metadata.json`.
- `session/start?track=…&description=…` — the per-session description.

Neither is required; both make a report readable months later.

10.3 Reading the Status

Each completed session carries an `overallStatus`:

Status	Meaning
`COMPLETED`	Every call's actual arguments matched the recording.
`COMPLETED_WITH_DEVIATIONS`	At least one call's input differed from the recording.

The companion endpoint `GET /service/recording/inputs?track={name}` returns just the recorded input sidecars, for comparing what a run *should* have sent.

Part II — Reference

Chapter 11. The `/service/*` HTTP Contract

`juke-remix-rest-service` exposes the control surface below. All endpoints accept `GET` unless noted; responses are `application/json` unless an attachment is streamed.

11.1 Recording Control

Endpoint	Description
`/service/record/start?track={name}`	Begin recording into the named track. Optional `&label={text}` sets the track label.
`/service/record/end`	Flush and close the ZIP; stream the bytes back as an attachment.

11.2 Replay Control

Endpoint	Description
`/service/replay/start?track={name}`	Switch to replay, pinned at the named track.
`/service/replay/disable`	Pass through without leaving replay mode.
`/service/replay/enable`	Re-enable replay after a `disable`.

11.3 Recordings Catalog

Endpoint	Description
`/service/recordings`	JSON array of every recording name the agent holds.
`/service/recordings/{name}`	Stream the named recording's ZIP bytes; 404 when unknown.

11.4 Sessions

Endpoint	Description

`/service/session/start?track={name}`	Start a per-session replay. Optional `&description={text}`. Sets the `JUKE_SESSION` cookie.
`/service/session/stop`	End the active session; clear the cookie.
`/service/session/status`	The active session's id, track, age, and step counter.
`/service/sessions`	Read-only view of every active session (`lastCall`, `percentComplete`).

11.5 Reports

Endpoint	Description
`/service/recording/report?track={name}`	Consolidated JSON report of every completed session for the track.
`/service/recording/inputs?track={name}`	The recorded input-argument sidecars for the track.

11.6 Coverage

Available only when `juke.coverage.enabled=true` (Chapter 9).

Endpoint	Description
`/service/coverage`	Combined summary — server and UI in one response; carries a top-level `passed` for one-call CI gates.
`/service/coverage/server`	Live server (JaCoCo) coverage summary; HTML at `/coverage/server/`.
`/service/coverage/ui`	Front-end (nyc/Istanbul) coverage summary; HTML at `/coverage/ui/`.

11.7 Remix

Endpoint	Description

`/service/remix/exceptionSchedule?classAndMethodSequenceId={id}&exceptionType={type}&exceptionMessage={msg}`	Schedule a one-shot exception on the named recording entry.
`/service/remix/delaySchedule?classAndMethodSequenceId={id}&delayTimeMs={ms}`	Schedule a delay before the named entry returns.
`/service/remix/clear`	Clear all scheduled delays and exceptions, so a fault injected in one replay run does not carry into the next.

11.8 Plugin Registry

Endpoint	Method	Description
`/service/plugins/register`	POST	Register a plugin.
`/service/plugins/{id}/heartbeat`	POST	Liveness ping; rotates the plugin token.
`/service/plugins/{id}/deregister`	POST	Voluntary teardown.
`/service/plugins`	GET	List registered plugins.
`/service/plugins/capabilities`	GET	Union of advertised capabilities.
`/service/plugins/{id}`	GET	Detail for one plugin.

Chapter 12. Configuration Reference

Juke reads layered YAML — `juke-defaults.yml` (in the jar) ← `application.yml` ← `application-{profile}.yml`. Every key has a `-Djuke.*` system-property alias; YAML is preferred for everything but one-off CI overrides.

12.1 Core Keys

Key	Default	Notes
`juke.enabled`	`false`	Master toggle. Nothing activates unless `true`.
`juke.path`	`${java.io.tmpdir}/juke`	Directory of recording ZIPs.
`juke.mode`	`ignore`	Global mode; usually flipped via REST.
`juke.zip`	*(none)*	Default ZIP name for global record/replay.
`juke.tests`	*(none)*	Whitelist of allowed track names.
`juke.disabled`	`false`	Soft kill-switch (`juke.enabled=false` is preferred).
`juke.args-validation`	`warn`	`off` / `warn` / `strict` (see §5.4).
`juke.storage.folder.path`	`recordings`	Folder root for the `JukeStorage` SPI.
`juke.storage.backend`	`folder`	`db` activates the Enterprise JPA backend.

12.2 Coverage Keys

Read by the optional `juke-coverage` module (Chapter 9).

Key	Default	Notes

`juke.coverage.enabled`	`false`	Opt-in toggle for the `/service/coverage/*` endpoints.
`juke.coverage.classes`	*(unset)*	Path to the application's `target/classes`.
`juke.coverage.sources`	*(unset)*	Path to `src/main/java` for HTML source highlighting.
`juke.coverage.report-dir`	`${user.home}/juke-demo/coverage/server`	Where the server HTML report is written.
`juke.coverage.ui-report-dir`	`${user.home}/juke-demo/coverage/ui`	Where the nyc UI report is read from.
`juke.coverage.bundle-name`	`Application under test`	Display name for the server coverage bundle.
`juke.coverage.threshold.server.line`	`0`	Minimum server line coverage (%). `0` = no gate.
`juke.coverage.threshold.server.branch`	`0`	Minimum server branch coverage (%).
`juke.coverage.threshold.server.instruction`	`0`	Minimum server instruction coverage (%).
`juke.coverage.threshold.ui.line`	`0`	Minimum UI line coverage (%).
`juke.coverage.threshold.ui.statements`	`0`	Minimum UI statement coverage (%).
`juke.coverage.threshold.ui.functions`	`0`	Minimum UI function coverage (%).
`juke.coverage.threshold.ui.branches`	`0`	Minimum UI branch coverage (%).

When any threshold is set, every coverage response carries `passed: false` the moment a metric falls short — see §9.6 for the CI-gate pattern.

12.3 The Per-Environment Pattern

```
# application.yml — base; no enabled key, so Juke is off juke: path: /tmp/juke-tracks #
application-dev.yml juke: enabled: true # application-prod.yml — omit juke.enabled entirely
```

Activate with `SPRING_PROFILES_ACTIVE=dev`. Juke turns on or stays invisible with no code change.

Chapter 13. The Storage SPI

`JukeStorage` is the framework's storage abstraction. A host can replace it without touching any other Juke code.

13.1 Two Abstraction Levels

- **Per-recording** — read and write entries within one ZIP (`readFromFile`, `writeToFile`, `getFileNames`, `path`).
- **Recording-store** — enumerate, load, and save whole recordings by name (`listRecordings`, `loadRecording`, `saveRecording`), used by the `/service/recordings` catalog.

13.2 The Default Implementation

`JukeZipDAOImpl` treats `juke.storage.folder.path` as the store and each `*.zip` file as a per-recording handle. It is registered by `JukeStorageAutoConfiguration` as `@Bean @ConditionalOnMissingBean`.

13.3 Custom Backends

Expose your own `JukeStorage` bean and the autoconfiguration steps aside. The Enterprise `juke-scenario-service` does exactly this: its `JukeJpaStorageBackend` activates with `juke.storage.backend=db` to keep recordings in a database. The agent code is identical either way.

Part III — Appendices

Appendix A. Sample Applications

The `juke-samples/` directory contains runnable references:

Sample	Demonstrates
`juke-sample-greeting`	The canonical app — one REST endpoint, one `@Juke` seam, a bundled React (Vite) SPA. The basis of *DEMO.md*.
`juke-sample-coverage`	The functional-coverage demo (Chapter 9). A live dashboard polls `/service/coverage` while you click through the journey; bundled `demo-start-server.{ps1,bat}` and `demo-run-playwright.{ps1,bat}` launchers wire up the JaCoCo agent and the nyc report path.
`juke-sample-todo`	A plain REST CRUD surface; the false-positive / `@JukeIgnorable` walk-through.
`juke-sample-session`	Per-session replay and the Playwright specs, including the visual demo.
`juke-sample-annotations`	`@Juke` on fields, methods, and constructor parameters.
`juke-sample-exceptions`	Exception and latency flows (Chapter 8). A SKU order places three orders through a `@Juke` OMS seam, driven four times — record, deterministic replay, replay with an injected delay ("queued"), and replay with an injected exception ("technical difficulties"). Confirmation numbers are `@JukeIgnorable`; coverage is shown in a separate popup.

`juke-samples/DEMO.md` is a 15-minute guided walk-through built on `juke-sample-greeting`.

Appendix B. Troubleshooting

Recording is empty or only contains `juke.json`. Confirm `juke.enabled: true`, that the global mode is `record`, that the `@Juke` bean is injected through Spring (not `new`-ed), and that the component scan covers `org.juke.framework`.

`JukeReplayNotFoundException`. The replay asked for a sequence the recording does not have — usually call-order drift between record and replay, or a `juke.zip` pointing at the wrong track. Re-record or fix the track name.

`INPUT MISMATCH` warnings. `juke.args-validation` found the live arguments differ from the recorded ones. Re-record, set the value to `off`, or set it to `strict` to fail loudly in CI.

`H2` / database errors at startup. A Community build is autoconfiguring JPA — `juke-scenario-service` or `spring-boot-starter-data-jpa` is on the classpath unintentionally. Check `mvn dependency:tree`.

Tracks bleed across Playwright workers. You are using `replay/start` for a process-wide flip instead of the per-session API. Switch to `/service/session/start` so each worker carries its own `JUKE_SESSION` cookie.

`/service/coverage/server` reports "agent not attached". The server was started without `-javaagent:jacoco-agent.jar`. Coverage degrades gracefully — attach the agent and retry.

Appendix C. Glossary

Track

A named recording — one ZIP archive of captured interactions.

Seam

The `@Juke`-annotated injection point where interception happens.

Mode

`record`, `replay`, `ignore`, `disable`, or `juke` — what the proxy does at the seam.

Session

A cookie-scoped, per-track replay context, isolated from other sessions on the same JVM.

Sequence

The 1-based per-`(type, method)` counter that orders recorded calls.

Sidecar

An `*.args.json` or `*.type.json` file accompanying a response in the ZIP.

Remix

Runtime fault injection — scheduled exceptions and delays on named entries.

Functional coverage

Coverage measured while real user-journey tests drive a Juke-replaying server, as distinct from unit-test coverage.

Colophon

The mascot of the *Juke Manual* is the **superb lyrebird** (*Menura novaehollandiae*), a ground-dwelling songbird of southeastern Australia. The lyrebird is among the finest vocal mimics in nature: it listens to the sounds of its environment — other birds, and sometimes mechanical noise — and reproduces them later with uncanny fidelity. A creature whose whole craft is *recording the real world and replaying it on demand* is a fitting mascot for Juke.

The cover image is a nineteenth-century engraving. The text is set in a serif body face with `Constant Width` for code.